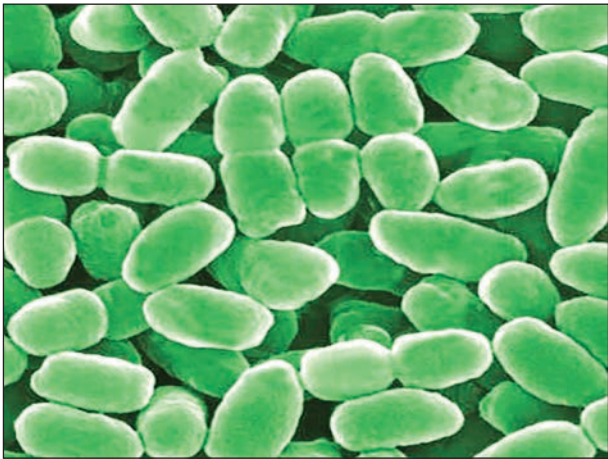


4. Whooping Cough is a disease caused by the bacterium *Bordetella pertussis* and it can be very serious or even fatal in young children and babies. In the UK, since 1950, young children and babies have been protected from Whooping Cough by vaccination and this reached the target of 95% of children by 1970. By this time there were very few cases.



Bordetella pertussis

In 1974, a medical journal published a report stating that the vaccine had damaged the nervous system in some children. Newspapers spread alarming information to parents throughout the UK and the percentage of children vaccinated fell to 30%.

**“Whooping Cough vaccine can
cause brain damage and death!”**

In 1978, the number of cases of Whooping Cough had risen to 2000 and by 2012 it was 7 730. The antibiotic drugs used to treat Whooping Cough were becoming ineffective.

The negative media reports were discredited and by 2017 the percentage of young children vaccinated in the UK had gone back up to 73%. The target remains at 95%.

Use the information to answer the questions.

- (a) (i) State the scientific name of the micro-organism which causes Whooping Cough. [1]

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- (ii) Complete the sentence, by choosing words from the list: [2]

antibodies antigens lymphocytes phagocytes

Micro-organisms which cause infections carry

The in the bloodstream then produce large numbers of

..... which can destroy the infecting micro-organisms.



(b) Choose the correct answer to complete the sentence. [1]

The percentage of children vaccinated against Whooping Cough decreased after 1974 because of newspaper articles which said

- A The vaccine had become less effective.
- B The vaccine had serious side effects.
- C The disease had become less serious.
- D Cases of the disease were reducing.

Answer

(c) (i) Calculate the percentage increase in cases of Whooping Cough between 1978 and 2012. [2]

Use the formula:

$$\frac{\text{number of cases in 2012} - \text{number of cases in 1978}}{\text{number of cases in 1978}} \times 100$$

Percentage increase =

(ii) State the evidence that the increase in the number of cases of Whooping Cough was linked to a reduction in the percentage of children being vaccinated. [2]

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Examiner
only

- (d) (i) By 2017, the number of cases of Whooping Cough had reduced. The NHS continued to encourage the vaccination of young children and pregnant women.

Use the information on page 11 to give **two** reasons why it was necessary to continue to encourage these vaccinations. [2]

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- (ii) In some countries the vaccination of children is compulsory. Suggest why some people object to compulsory vaccination. [1]

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.....

- (e) The following six sentences explain why antibiotics became ineffective for treating Whooping Cough, but they are not in the correct order.

- 1 Many doctors over-used the antibiotics.
- 2 The resistant bacteria multiplied rapidly.
- 3 The antibiotics could no longer be used to kill the bacteria.
- 4 Almost all the bacteria became resistant to the antibiotics.
- 5 Doctors discovered antibiotics which killed Whooping Cough bacteria.
- 6 Some bacteria survived because they were resistant to the antibiotics.

Complete the list below to put the six sentences in the correct order by writing the missing numbers on the dotted lines. [2]

5

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6

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4

.....

13

